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ORIGINAL RESEARCH

Preparation of methylated TEMPO-oxidized cellulose nanofibril hydrogel with high-temperature sensitivity

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Abstract TEMPO-oxidized cellulose nanofibril (TOCNF) hydrogel presents unique advantageous characteristics and has been envisioned as a building block to design novel biobased materials. However, the failure to maintain the gel shape and rapid decline of viscosity at elevated temperature (> 120 °C) limit the application of TOCNF hydrogel in high temperature environment of oil fields. In this study, dimethyl carbonate (DMC), a green methylation reagent, was employed to introduce methyl groups into TOCNFs to endow the DMC-TOCNF hydrogel with improved resistance at higher temperatures. The DMC-TOCNFs and original TOCNFs were characterized by fourier transform infrared (FTIR), X-ray photoelectron spectroscopy, atomic force microscopy, thermogravimetric analysis and rheological tests. The results show that the degree of substitution of DMC-TOCNFs was 0.078, 0.390 and 0.630, respectively. DMC-TOCNFs displayed the characteristics of gelation at high temperature, which increased with higher degree of substitution (DS). Of particular importance, the viscosity of DMC-TOCNFs dramatically raised from 226 to 2.9 × 104 mPa s when the temperature rose from 30 to 160 °C at a DS of 0.630. The methylated-TOCNF hydrogels displayed great potentials in expanding the application of TOCNFs at high temperatures.

Keywords TEMPO-oxidized cellulose nanofibril · Dimethyl carbonate · Thermal behavior · High temperature sensitivity

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Introduction

Cellulose nanofibrils (CNFs), w, can be isolated from renewable raw materials such as northern bleached softwood kraft (NBSK) via mechanical, thermal, or enzymatic processes (Malladi et al. 2018; Zhuo et al. 2017). TEMPO-oxidized CNFs (TOCNFs), prepared from wood pulps through the sequential process of high pressure homogenization, have been widely employed in various applications such as rheology modifiers, thickeners, reinforcing materials, and coating components, owing to their special properties such as 3-dimensional network structure even at low concentration (Abe et al.2007; Saito et al. 2007). For example, Fanny et al. evaluated the suitability of

CNFs as a thickening agent to provide high viscosity silver nanowires screen printing ink (Hoeng et al. 2017). Of particular interest is the utilization of TOCNFs as a good thickening material in oil and drilling industry (Xla et al. 2019). Relative to traditional polymers, CNFs have additional advantages including natural renewability, biodegradability, nanometer size, and ease of surface modification. The high temperature in the oil well environment stipulates requires oilfield chemicals with higher thermal stability (Liu et al. 2019), however, the viscosity of TOCNF hydrogel loses quickly at high temperatures since the surface of TOCNFs only contains carboxyl groups, thus urging the improvement of the thermal stability of the gel to expand its application at high temperatures.

Higher thermal stability could be obtained by improving the hydrophobic properties of polymers. Hydrophobic association can effectively maintain the viscosity of oilfield polymers by entangling and increasing the viscosity of polymer solutions (Candau et al. 1994). For example, Liu et al. grafted butyl acrylate (BA) onto TOCNFs backbone through the ceric ammonium nitrate-induced radical polymerization system to prepare a type of modified TOCNFs with high thermal stability (Xiongli et al. 2019). Zhu et al. introduced an amphiphilic polymer with twin-tailed hydrophobic groups that showed reinforced performance on temperature resistance (Zhu et al. 2017). Meanwhile, the amphiphilic polymer solution was thickened as a result of the strength of the hydrophobic association increased (Zhou et al. 2017). Methylation modification is one of the typical hydrophobic modification methods to improve the thermal stability, for instance, the methylated cellulose (MC) was reported to present thermal property (Bajwa et al. 2009; Hirrien and Ross-Murphy 2000).

Methyl chloride, methyl iodine and dimethyl sulfate were often selected as the methylating agents while their toxicities are not viewed favorably for practical applications. In contrast, dimethyl carbonate (DMC), known as a methylating agent for phenols (Barcelo et al. 1990; Lee and Shimizu 1998) and anilines (Selva et al. 2001) that does not produce inorganic salts, can be applied as an environmentally friendly alternative (Shieh et al. 2010), only generating byproducts methanol and carbon dioxide (Lui et al. 2016). However, to the best of our knowledge, limited research on the O-methylation of cellulose with DMC was reported. For example, methyl carbonate cellulose was synthesized with DMC in [P_{8881}][OAc]/DMSO using IL (Ionic Liquid) as a secondary solvent and catalyst (Labafzadeh et al. 2015). Hou et al. described a novel approach for the methylation reaction of cassava starches by employing thiocarbamide as a catalyst under microwave irradiation wherein DMC was used as the methylating reagent and solvent (Chengmin et al. 2011). Several innovative and green processes using DMC to produce CNFs and cellulose derivatives were reported by Ramzi (Khiari et al. 2019) and Khiari et al., while the effect of DMC on the temperature resistance, high thermal stability and the hydrogel characteristics of modified TOCNFs were not yet studied (Hodges et al. 1979b; Khiari et al. 2017).

In this work, TOCNF hydrogel was prepared from softwood kraft pulp using a two-step process using TEMPO-mediated oxidation to introduce anionic charges, followed by high pressure homogenization methylation of the TOCNFs with DMC to graft methyl groups. Subsequently, Fourier transform infrared (FTIR) spectroscopy, nuclear magnetic resonance (NMR), X-ray photoelectron spectroscopy (XPS), atomic force microscopy (AFM), thermogravimetric analysis (TGA) and rheological analysis were conducted to characterize the chemical, microstructure, thermal and rheological properties of the obtained DMC-TOCNFs. The main purpose of this study was to fabricate TOCNF hydrogels with high thermal stability for the oil and drilling industry as a thickening agent.

Experimental

Materials

TEMPO-oxidized cellulose nanofibrils were provided by Tianjin Woodelf Biotechnology Co. Ltd. (Tianjin, China), the carboxyl content of TOCNFs was 1.05 mmol/g, TOCNFs were 5–10 μm in length and 10–20 nm in diameter. The component of TOCNFs does contain 7.61% hemicellulose, and no lignin exist, and the surface charge was −43.0 mV. Analytical grade cetyltrimethylammonium bromide (CTAB, 99.0 wt%), N, N-dimethylacetamide (DMAc, 99.5 wt%) was purchased from Sigma-Aldrich Chemical Co. Ltd. (Tianjin, China). Dimethyl carbonate (DMC, 99.0 wt%, analytical grade) was obtained from Shanghai Yi Shi Chemical Co. Ltd. (Shanghai, China).

Methylation of TOCNFs

Firstly, TOCNFs (20 g) were fully dispersible in DMAc (80 g). CTAB (0.4 g) and Na_2CO_3 (12 g) were dissolved in 10 mL of ethanol and distilled water, respectively, then these solutions and DMC were added into the resultant TOCNFs/DMAc suspension. The mixture was heated for 10 h at 73 °C then the crude mixture was filtered and the solid cake was rinsed with ethanol (75%, v/v) and deionized water three times (Yufang et al. 2010). Weight recovery ratio of DMC-TOCNFs after methylated was calculated using the dry mass of the sample before and after the reaction. The weight recovery ratio was 98%.

Analytic methods

The determination of the degree of substitution (DS) was determined following the method described by Zeisel (Hodges et al. 1979a) based on the reaction between the sample and the hydroiodic acid to form methyl iodide.

$$R{-}OCH_3 + HI \rightarrow ROH + CH_3I$$

$$CH_3I + AgNO_3 \rightarrow AgI + CH_3ONO_2$$

The DS was calculated by Eq. 1:

$$DS = \frac{162M}{31 - 14M} \tag{1}$$

where M is the weight percentage of the methyl groups.

Fourier transform infrared (FTIR) analysis of the samples was conducted with a FTIR-650 spectrometer (Guangdong Co. Ltd., Tianjin; China) in the wavenumber range of 4000–500 cm^{-1} with a resolution of 4 cm^{-1} for 32 scans to record the modifications in the intensities of the characteristic bands before and after methylation. The original TOCNF and DMC-TOCNF samples were dried at 105 ± 2 °C for 4 h to a constant weight to determine the moisture content. The samples were prepared using a mixture of 0.001 g of sample and 0.1 g of KBr.

X-ray photoelectron spectroscopy (XPS) experiments was conducted on an ESCALAB 250Xi XPS microprobe (Thermo Fisher Scientific, USA) with radiation (0–5000 eV). Spectrum NT and the CAH signal were used to deconvolute the peaks.

The solid-state ^{13}C nuclear magnetic resonance (^{13}C NMR) experiments were carried out with a JNMECZ600R 600 spectrometer (JEOL Ltd., Japan) at 125 MHz under a magic-angle spinning speed of 8 kHz. The contact time for cross-polarization and the recycle time were 1.6 ms and 12 s, respectively.

Thermal stabilities of the samples were tested using a TGA Q500 (TA Instruments, USA) instrument at the temperature ranging from room temperature to 620 °C under continuous nitrogen flow of 70 mL/min with a heating rate of 10 °C/min.

The morphological characteristics of TOCNFs and DMC-TOCNFs were observed by atomic force microscopy (AFM, JEOL 2010, Japan) using a silicon oxide cantilever probe at 1 Hz scan rate at 600 nm/s by dropping about 0.3 mL of fully-dispersed sample on the mica plate and drying for 9 h at room temperature.

Rheological analysis

Rheological analysis was performed with an Anton Paar MCR 302 rheometer (Germany) equipped with a CC27 measuring system (Measuring Bob, D = 26.6 mm, L = 40.0 mm; Cup, D = 28.9 mm); the viscosities of the samples (1 wt%) were measured at a shear rate of 7.34 s^{-1}. The sample was held for 5 min before each experiment to attain thermal equilibrium. A thermal program used for heating the sample exerted a 0.5 °C/min rate from 30 °C to the gel temperature point. Subsequently, the sample was held at constant temperature for 30 min. All dynamic measurements were acquired in the linear range along the temperature range from 30 °C to the gel temperature point with heat-cool ramps. Then, tests were carried out at a constant frequency of 0.5 Hz with a constant strain of 1%.

Small amplitude oscillatory was performed with this rheometer to measure the elastic (G′) and viscous (G″) moduli (at 30 °C and gel temperature point) of these samples as a function of angular frequency from 0.1 to 100 Hz. An oscillation strain sweep was initially conducted to distinguish the linear viscoelastic region of the samples. The samples were diluted to 1 wt% with deionized water and then used for all the tests. Viscosity measurements were performed in triplicates with the same rheometer with a shear rate ranged from 0.1 to 1000 s^{-1}.

Results and discussion

DMC-TOCNFs preparation mechanism and degree of substitution (DS)

Figure 1 shows the schematic diagram of the CTAB-catalyzed modification of TOCNFs with DMC, in which DMC was activated by CTAB to methylate the hydroxyl groups of TOCNF then decomposed into methanol and CO_2. DMC-TOCNFs were gelated at high temperatures because of the hydrophobic associations between methyl groups (Hirrien et al. 1998).

To evaluate the effect of methyl group content on the properties of products, DMC-TOCNFs with different methyl group contents were produced by varying the DMC loadings during the modified process, starting with TOCNFs using a carboxyl content of 1.05 mmol/g. Figure 2 depicts the effect of DMC loading on the content of methyl group and DS of DMC-TOCNFs. As expected, the higher

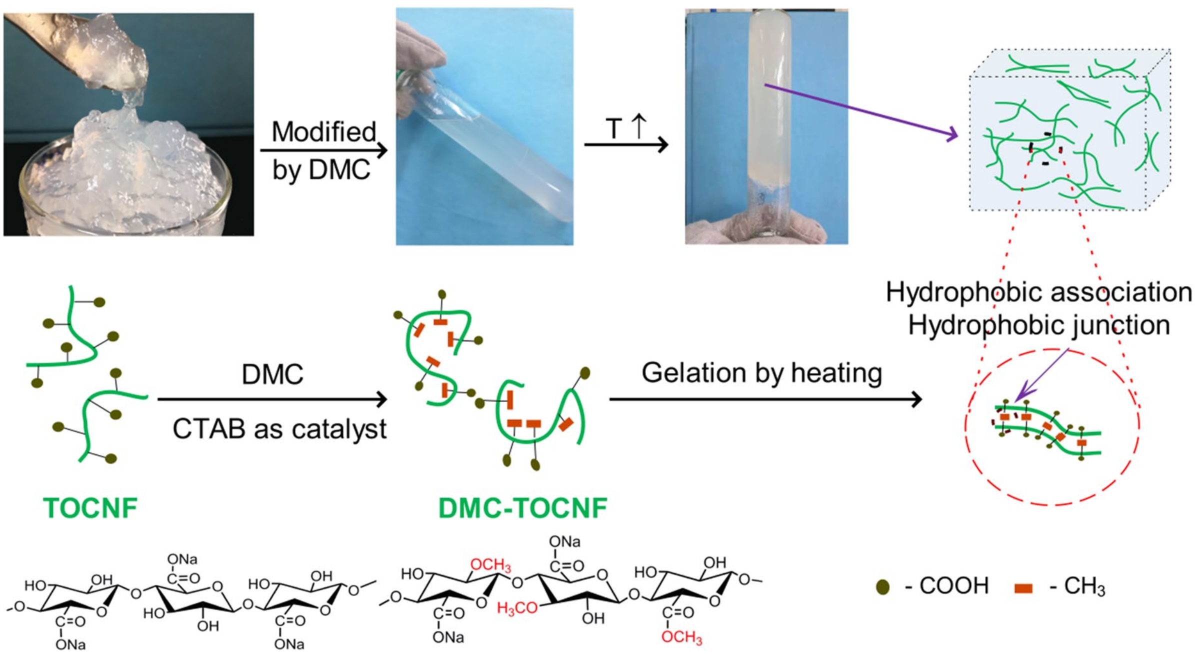

Fig. 1 Schematic diagram of DMC modified TOCNFs

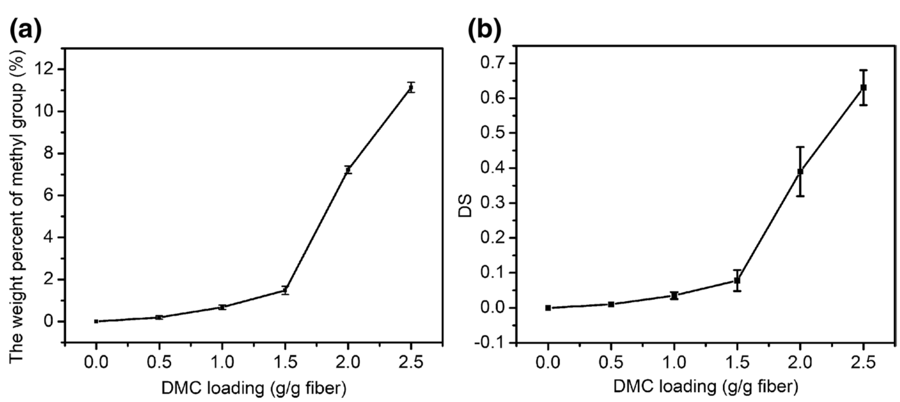

Fig. 2 **a** The effect of DMC loading on the content of methyl group of DMC-TOCNFs; **b** the methyl DS of DMC-TOCNFs

the DMC loading, the higher the content of the methyl groups and DS, evidenced by the significant improvement of the methyl group weight percent from 0.19% to 11.14% when raising the DMC loading from 0.5 to 2.5 g/g fiber. This is because when the dosage of DMC was 1.5 g/g or lower, a significant amount of DMC will lose after activation, and only a few mount of DMC could participate in the methylation reaction. Though the loss is inevitable, increasement of DMC dosage above 1.5 g/g can apparently improve the reaction rate, as more DMC was available in the system. Subsequently, the obtained DS was 0.010, 0.034, 0.078, 0.390 and 0.630, respectively. However, DS values of 0.010, 0.034 and 0.078 showed almost the same properties, and only 0.078, 0.390 and 0.630 were used for subsequent characterization.

Chemical characterization of DMC-TOCNFs

FTIR was conducted to confirm the corresponding changes of the molecular structure of TOCNFs after methylated modifications. The profiles of the spectra of TOCNFs and DMC-TOCNFs (0.390) were pretty identical, suggesting that the inherent structure of cellulose was not altered (Fig. 3a). The absorption bands at 3400 and 2900 cm^{-1} were assigned to O–H and C–H stretching vibration, respectively (Hu et al. 2014; Kim et al. 2013; Mei et al. 2016). The absorption peak at 1640 cm^{-1} was associated to the adsorbed water (Li et al. 2016), and the peak at around 1730 cm^{-1} corresponded to the carboxyl groups (Habibi et al. 2006). In addition, the enhancement of the intensity of peaks at 1450 cm^{-1} and 2850 cm^{-1} were caused by the –CH_3 stretching vibration (Varghese et al. 2007) and C–H stretching (Oliveira et al. 2015) in the grafted methoxy groups, respectively.

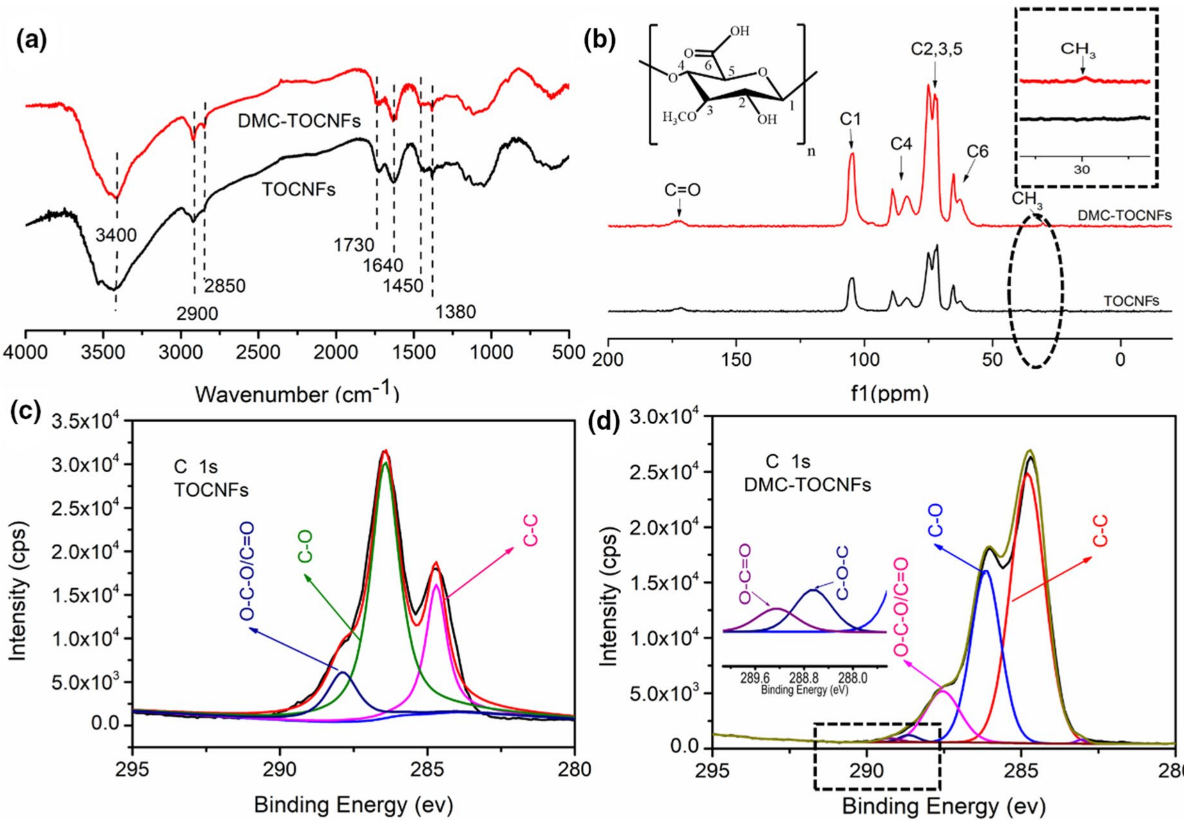

Fig. 3 **a** FT-IR spectra of TOCNFs and DMC-TOCNFs (0.390), **c** ^{13}C NMR spectra of TOCNFs and DMC-TOCNFs (0.390), and **b** Deconvoluted C 1s XPS spectra of TOCNFs and **d** DMC-TOCNFs (0.390).

The ^{13}C NMR spectra of unmodified TOCNF and DMC-TOCNF samples are shown in Fig. 3c. The characteristic cellulose signals (C1 104.7 ppm; C2, C3, C5 72–74 ppm; C4 83–89 ppm; C6 64.8 ppm) of TOCNFs could be easily assigned. The peak at 172.8 ppm was assigned to C=O, (Albert et al. 2004; Ma et al. 2018) and the new signal in DMC-TOCNFs located at 30.4 ppm corresponded to the methyl group (Sang et al. 1985). The absence of free DMC characteristic peaks at 54 ppm for CH_3 and 155 ppm for –O–(C=O)–O carbonate groups, which confirmed the absence of DMC in the samples.

The deconvoluted C 1s signals of TOCNF and DMC-TOCNF samples are shown in Fig. 3b, d, in which the deconvoluted C 1s signals of TOCNFs were fitted into three peaks: C–C at 284.7 eV, C–O at 286.7 eV, O–C–O and C = O at 287.8 eV (Niu et al. 2018), respectively. The appearance of the new peak of the C 1s signals in DMC-TOCNFs at 288.6 eV (Duan et al. 2019; Liu et al. 2020) is associated with the C–O–C contribution arising from methyl in DMC-TOCNFs, which again, confirmed the successful etherification of TOCNFs by DMC.

Characterization of DMC-TOCNFs

TGA and DTG were used to illustrate the thermal stability of TOCNFs and DMC-TOCNFs (Fig. 4), in which all samples presented almost identical profiles, evidenced by the similar $T_{5\%}$ (temperature corresponding to 5% weight loss), $T_{30\%}$ (temperature corresponding to 30% weight loss) and T_{max} (maximum degradation temperature) as shown in Table 1, demonstrating that the modification process did not change the thermal stability of the resulting derivatives.

The microstructures of all DMC-TOCNF samples are quite identical to that of TOCNFs (Fig. 5), despite the differences in DS, which illustrating that the methylation process did not significantly alter the size and structure of the resulting derivatives.

Rheological analysis

The rheological behaviors of DMC-TOCNFs and TOCNFs were studied as a function of temperature (Fig. 6a) and time (Fig. 6b). While the viscosity of TOCNFs was reducing with elevated temperature

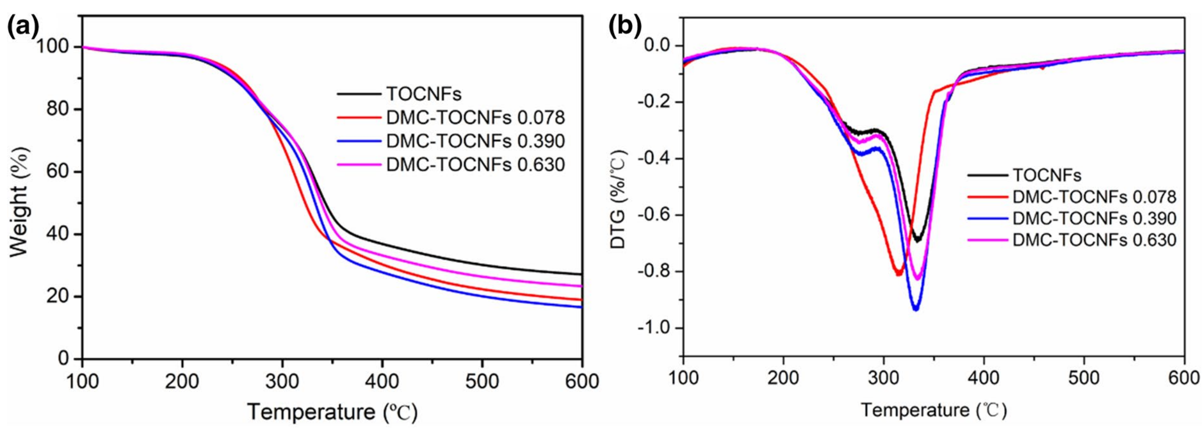

Fig. 4 Thermal decomposition profile of the control TOCNFs and DMC-TOCNFs. **a** TGA thermograms; **b** derivative thermal gravimetry

Table 1 Different degradation temperatures and the final residue rate of control TOCNFs and DMC-TOCNFs

Sample	$T_{5\%}$ (°C)	$T_{30\%}$ (°C)	T_{max} (°C)	Residue rate (%)
TOCNFs	224.1	311.2	333.1	27.1
DMC-TOCNFs 0.078	232.2	298.3	317.3	18.9
DMC-TOCNFs 0.390	227.2	305.1	332.2	16.6
DMC-TOCNFs 0.630	230.4	311.4	333.4	23.4

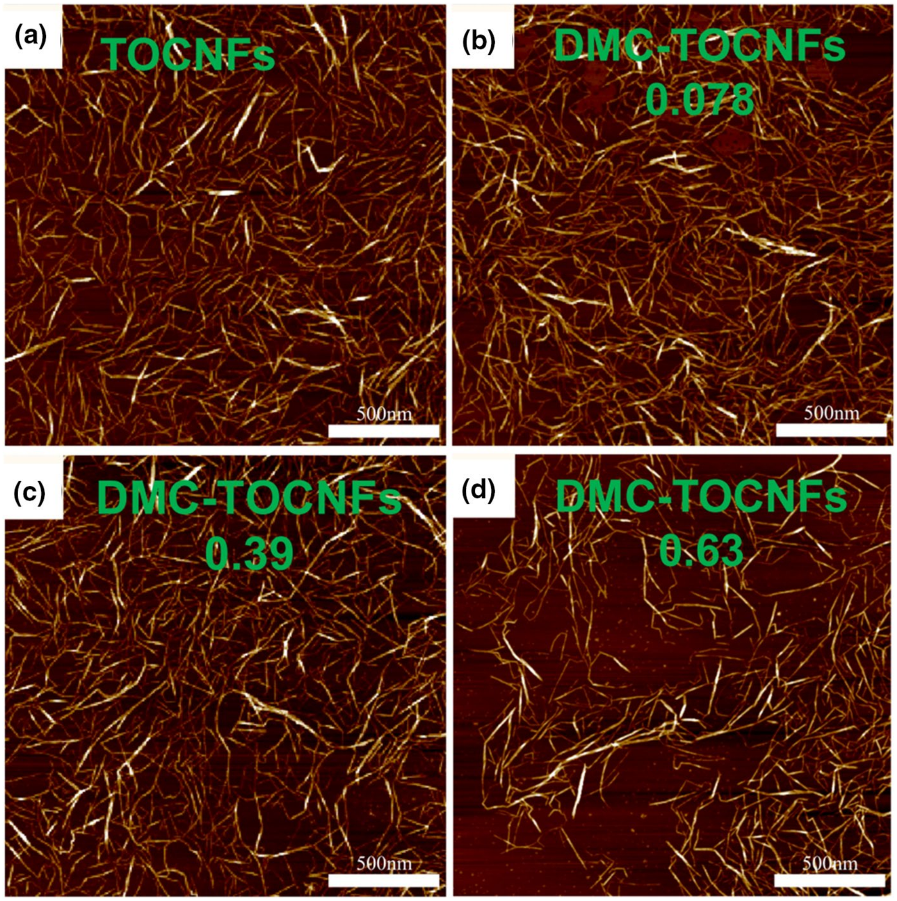

Fig. 5 AFM images of different samples. **a** TOCNFs; **b–d** DMC-TOCNFs

(Fig. 6a), declining from 2369 to 167 mPa s (Fig. 6a, black trace), likely due to the rapture of the hydrogen bonds between the carboxyl groups and the water molecules during heating process. During the time course study, its viscosity remained stable at constant temperature (130 °C), regardless of the testing time (Fig. 6b). In contrast, the viscosity of DMC-TOCNFs increased dramatically upon heating (Fig. 6a). For example, the viscosity of DMC-TOCNFs (0.630) increased from 226 to 2.9×10^4 mPa s, and such high viscosity revealed that DMC-TOCNFs have hydrogel properties at high temperatures. Though the hydrogen bonds between carboxyl groups and water molecules were destroyed upon heating (Li et al. 2002), the hydrophobic association of methyl groups in DMC-TOCNFs increased at elevated temperatures, which was beneficial for the inter-bonding of the methyl groups between DMC-TOCNFs. In addition, the self-assembly between DMC-TOCNF chains increased the diameter of nanofibers to present a gel state (Morozova 2020), as showed in Fig. 6d. In contrast to MC, DMC-TOCNFs have more carboxyl groups thus the DMC-TOCNF's gelation temperature increased with higher DS (Fig. 6c), in which the higher the MC substitution degree was, the lower the gel temperature point (Nasatto et al. 2015) would be.

Viscoelastic properties can be used to characterize the change of solution morphology. Storage (elastic) modulus G′ and loss (viscous) modulus G″, the main parameters represent the portion of the oscillation energy that is stored elastically and the energy dissipated by the system, respectively (Norinaga et al. 2002), were routinely measured to evaluate the sample in terms of viscoelastic properties (Yang et al. 2017). Therefore, the plot of the storage modulus G' and the loss modulus G" is another way to analyze the gelation performance of DMC-TOCNFs. Figure 7a shows the viscoelastic properties of TOCNFs and DMC-TOCNFs as a function of frequency at 30 °C. There was almost no difference in G′ and G″ between

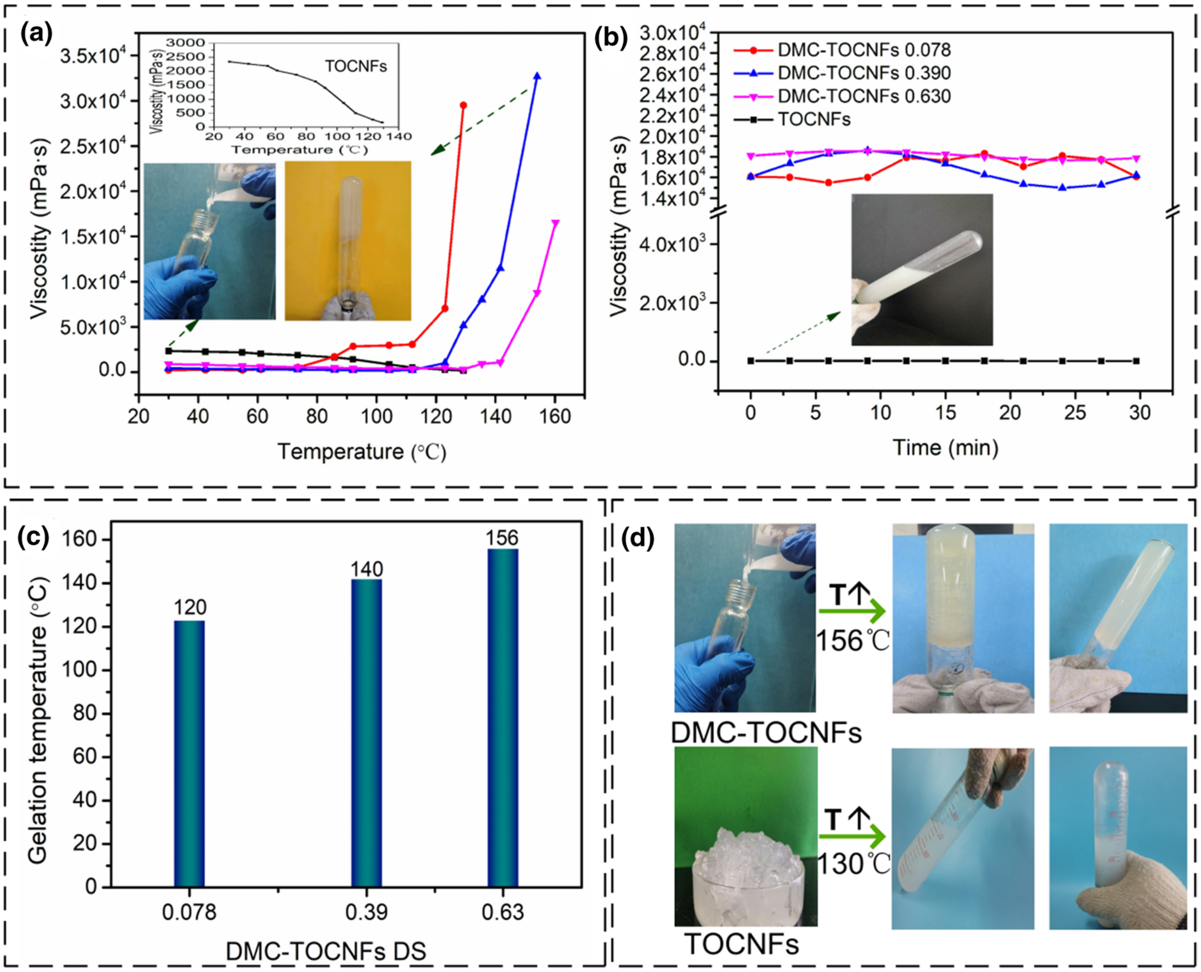

Fig. 6 **a** Viscosity of the TOCNFs and DMC-TOCNFs as a function of temperature at 7.34 s^{-1}; **b** The viscosity of TOCNFs and DMC-TOCNFs changes with time at the gel temperature point; **c** Gel temperatures for samples with different DS; **d** Appearance of DMC-TOCNFs (0.630) and TOCNFs before and after heating

TOCNFs and DMC-TOCNFs, indicating that viscous component became the dominating factor (Xia et al. 2007). After gelation (Fig. 7b), there was still little difference between the G′ and G″ of TOCNFs, implying that there was no phase change. In contrast, the G′ of DMC-TOCNFs was higher than their G″, suggesting the gel behavior of the DMC-TOCNFs at high temperature. A very pronounced shear thinning behavior was observed in the test of the viscosity of TOCNFs and DMC-TOCNFs as functions of viscosity shear rate (Fig. 7c), implying non-Newtonian behavior of these materials. Figure 7d shows the viscosity of TOCNFs and DMC-TOCNFs as a function of viscosity shear rate after gelation, in which the viscosities of DMC-TOCNFs were much higher than that of TOCNFs, again illustrating that DMC-TOCNFs have the properties of hydrogel at high temperature.

Conclusions

In this study, we reported the viability of producing DMC-TOCNFs from TOCNFs using DMC as an alkylating agent in heterogeneous conditions. The results showed that the raise of DMC loading increased the methyl group content of the resultant DMC-TOCNFs, and did not change the thermal stability and

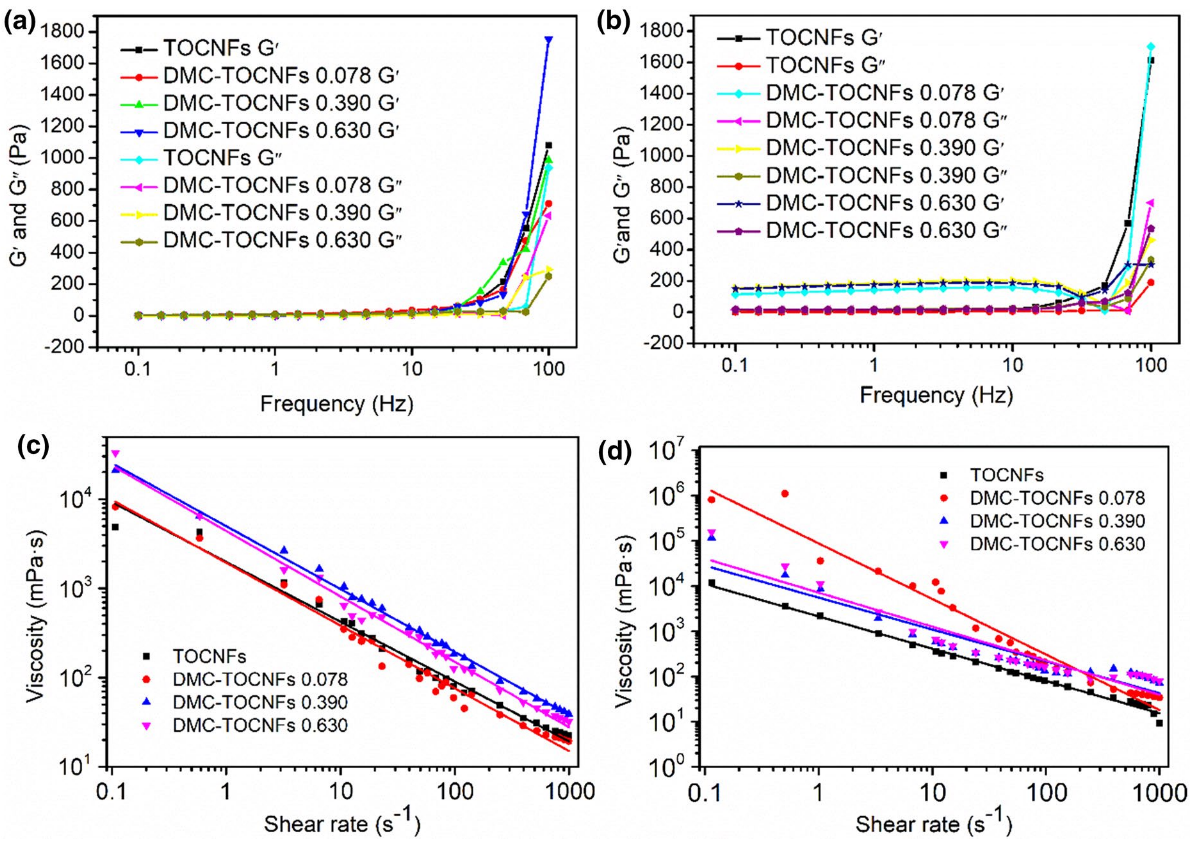

Fig. 7 **a** The viscoelastic properties as a function of frequency between the TOCNFs and DMC-TOCNFs at 30 °C; **b** The viscoelastic properties as a function of frequency between the TOCNFs and DMC-TOCNFs after gelation; **c** The viscosity of TOCNFs and DMC-TOCNFs as functions of viscosity shear rate; **d** The viscosity of TOCNFs and DMC-TOCNFs as functions of viscosity shear rate after gelation

structure of DMC-TOCNFs. Compared to un-modified TOCNFs, the resultant DMC-TOCNF hydrogel (1 wt%), with a degree of substitution of 0.630, presented the characteristics of gelling at high temperature, and maintained at 2.910^4 mPa·s at 160 °C. The increased high temperature sensitivity of the DMC-TOCNFs may lead to the expansion of the potential high-value applications in oil recovery thickening and other fields.

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